خدمتكم طريق خضناه لرضى الله

2026

حلول التجارب

مختبر الكيمياء العامة



لجنة السنافر

١٥/١٠
٥/١٠

ALBALQA APPLIED UNIVERSITY

FACULTY OF ENGINEERING TECHNOLOGYS

LABORATORY

EXPERIMENT « 1 »

Safety

NAME: [redacted]

REG.NUMBER: [redacted]

INSTRUCTOR: دكتور علي الشوابكة

LAB TIME: 2:30 - 5:30

DATE: 8/3/2026

Scanned with CamScanner

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Experiment (1)

* Safety Rules :-

- Introduces general laboratory safety rules.
- Emphasizes understanding instruction befor starting and asking questions when unsure.
- Highlights the impartance of knowing the location and use of safety equipment.

* Laboratory safety Equipment and Rules :-

- Explains the use of safety shower:, eye wash stations and first aid kits
- Lists general rules :-

① wear Lab coats and goggles

② No eating or drinking

③ No phones

- Report accidents immediately to the instructor
- Treat all chemicals as dangerous and wear gloves
- Keep chemicals in proper places

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Chemical Handling and Hazard Symbols:-

- Read labels carefully before using chemicals.
- Never taste or smell chemicals
- Avoid contact with eyes or skin, rinse immediately if exposed
- Learn hazard pictograms such as flammable, toxic corrosive, oxidizing, harmful, radioactive environmental hazard.
- Take small amounts of chemicals and never return unused portions to stock bottles.

common Laboratory Apparatus - Balances:-

- Never pour water into acid, always pour acid into water.
- Top-loading balance has accuracy ± 0.01
- Analytical balance has accuracy ± 0.0001
- use suitable containers, tare the balance, transfer chemicals carefully
- clean spills immediately and record reading

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Laboratory Glassware (part 1)

- Beaker: reaction container, mixing, heating, ~~Erlenmeyer~~
- Erlenmeyer flask: mixing, titration
- Volumetric flask: preparing solutions of known concentration
- Graduated cylinder: measuring liquid volumes
- pipette: dispensing small liquid quantities

Laboratory Glassware (part 2)

- Burette: precise liquid addition, titration
- Round bottom flask: heating liquids
- Desiccator: protecting hygroscopic chemicals
- Test tube: holding or mixing small samples
- Funnel: transferring liquids or filtration
- Glass rod: stirring liquids

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Bunsen Burner:-

Diagram and explanation of Bunsen burner parts
Adjust flame for clean combustion (blue flame)
Never heat flammable chemicals directly
Keep area clear of papers and combustible materials Never leave flame unattended
Shut off gas after use.

Other common Laboratory Apparatus:-

Crucible: heating chemicals to high temperatures
Wash bottle: rinsing glassware
Fire gauze: protecting glass ware during heating
Spatula: collecting small quantities of chemicals

(10/10 26/3)

Al-Balqa Applied University

Faculty of Engineering Technology

Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 2

Physical Properties: part 1 (Density and Solubility)

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 15 - 3 - 2026 [2:30 - 5:30]

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(2) تجربة

Data & Calculation:

Solubility:

*The table below for testing solute with solvents:

Solute / Solvent	NaCl	I_2	Glucose	Hexane [C_6H_{12}]
H_2O	S	N	S	N
Hexane [C_6H_{12}]	N	S	N	—

Part A: the density of an irregular shaped solid: "Lap" * في التادم

Unknown NO.: Zn

Mass of the solid	12.14 g
Initial volume of water	10 cm
Final volume of water	11 cm
Volume of solid	1 cm
Density of solid	12.14 g/ Cm^3

$\Delta = \frac{m}{V}$

Show Calculation:

To calculate the Density we use $\Delta = \frac{m}{V} \Rightarrow \frac{12.14}{1} =$

$12.14\ g/cm^2$

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Part B: density of unknown liquid.

Unknown NO.: _____

Mass of empty volumetric flask / Pycnometer	27.629 g
Mass of volumetric flask / Pycnometer + liquid	78.6719
Mass the unknown liquid	51.042 ⇒ $m_f - m_i$
Volume of unknown liquid	25mL
Density of liquid	2.04168 g/mL ⇒ $\frac{m_{tot}}{V_{tot}}$

Show Calculation:

to calculate the density we use $\Delta = \frac{m}{V}$

to calculate m_{liquid} we use $m_f - m_i$

notes:- for (1)
K_2SO_4: is an ionic compound that ions stabilized by H_2O molecules (ion-dipole forces)
C_2H_5OH: is polar molecules capable of forming hydrogen bonds with H_2O

Questions

1) Which of the following compounds is water soluble? Why?

A) K_2SO_4 → S
B) C_2H_5OH → S
C) $CH_3(CH_2)_{10}CH_3$ → N
D) $CaCO_3$ → N

*Both of K_2SO_4 and C_2H_5OH are water Soluble this follows the "Like Dissolves Like" principle. refar the note↑

2) In which of the following solvents KCL is dissolve? Explain your answer.

A) H_2SO_4 → ionic solvent
(B) HCl
C) C_6H_6 (Benzene) → non-polar solvent

Explanation:

Hcl ⇒ (in Solution) provides a polar environment that allows for (ion-dipole) interactions, leading to dissolution

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3) What is the affect if you introduced a large quantity of compound into test tube during the experiment of solubility?

- Introducing an excessive amount of a compound will likely result in a "Saturated Solution"

- The Explanation of this :-
every Solvent have a specific limit to how much solute it can dissolve at a given temperature. Once this limit is reached, the Solution becomes saturated. Any additional compound introduced by and this point will not dissolve and will instead settle at the bottom of the test tube as a precipitate (undissolved Solid)

4) Will the measured density of a solid material be affected if a part of the solid dissolved inthe liquid used and the volume increased? Explanation:

- yes, because when the volume increase the density will decrease.

$\Delta = \frac{m}{v} \Rightarrow \Delta \propto \frac{1}{v}$

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Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 3 Physical Properties: part 2 (Melting and Boiling Points)

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 22/3/2026

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تجربة (3)

Data & Results:

The Melting point of unknown: ----------

The Boiling point of unknown: 61.2 C - $CHCl_3$ (chloroform)

Questions

1) What is the effect of each following factors on the measured melting point of a solid (increase, decrease, has no affect) & give good explanation for each answer?

a- Rapid heating: Decrease

- The sample doesn't reach thermal balance, so it looks like it melts earlier

b- A large quantity of compound was introduced into the capillary tube: Increase

- A large amount takes more time for heat to reach the center, leading to higher observed melting point and a wider melting range

c- Impurities: Decrease

- Impurities disrupt the crystalline lattice of the solid, weakening the bonds and lowering the energy needed to melt it.

2) What is the effect of each following factors on the measured boiling point of a liquid(increase, decrease, has no affect)?

a- If the test tube base and the thermometer bulb are not at the same level:

- Increase → If the thermometer is lower than the test tube.
- Decrease → If the thermometer is higher than the test tube.

b- If there's some glass shards with liquid sample:

- has no effect

c- If the atmospheric pressure increase:

- Increase

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General chemistry lab 1

Course Number: 30206111

Experiment No. 5

Molecular Weight of a VolatilLiquids

(Dumas Method)

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 12-4-2026

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تجربة (1)

Data and Calculations:

DATA	Values
Mass of empty flask + aluminum foil + rubber band (m_1)	51.86
Mass of empty flask + aluminum foil + rubber band + condensed vapor (m_2)	52.16
Mass of condensed vapor (m_{gas})	0.30
T (°C)	88
T (K) + 273	361
Pressure (mmHg)	792
Pressure (atm)	1.04
Volume of flask (mL)	64
Volume of flask (L)	0.064
Molecular weight of volatile liquid (g/mol)	133.9 ≈ 134 g/mol

Show Calculations

$mass_{gas} = m_2 - m_1 \Rightarrow 52.16 - 51.86 = 0.3\ g$

$Pressure\ (atm) = \frac{792}{760} = 1.04210\ atm$

$1L \longrightarrow 1000ml \Rightarrow X = \frac{64}{1000} \Rightarrow 0.064\ L$

$X \longrightarrow 64$

$M_{mass} = \frac{m \cdot R \cdot T}{P \cdot V} \Rightarrow \frac{0.3\,(0.08206)(361)}{(1.04)(0.064)} = 133.5 \approx 134\ g/mol$

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Questions:

Q1- A sample of an unknown liquid is vaporized in a flask having a volume of 265 mL at 99.9°C. The vapor has a mass = 0.725 g, and exerts a pressure = 745 torr. Calculate the gram molecularweight of the unknown liquid.

$M_{mass} = \frac{m \cdot R \cdot T}{P \cdot V} \Rightarrow \frac{(0.725)(0.08206)(372.9)}{(0.98)(0.265)} = 85.4\ g/mol$

note:-

$\Rightarrow P_{atm} = \frac{P_{torr}}{760} = \frac{745}{760} \Rightarrow 0.98\ atm$

$\Rightarrow T_K = 99.9 + 273 \Rightarrow 372.9\ K$

Q2- Suggest 3 possible sources of experimental error in this experiment.

- Error (1): wrong total evaporate: $m_{gas} \uparrow$ $M_m \uparrow$
- Error (2): wrong pressure: $P \uparrow$ $M_m \uparrow$
- Error (3): wrong volum in flask: $V \downarrow$ $M_m \uparrow$

Q3- A student performing this experiment accidently tore the foil cap, so that extra vapor escaped from the flask before it was removed from the heat. Describe how this would affect the mass of vapor and the calculated molar mass.

Mass of vapor: The measured mass of the vapor willbe lower than the actual value this is because the torn foil cap allowed some for the vapor to escape from the flask befor it could be cooled and weighed

calculated molar mass: will be lower. reason: according to $M = \frac{mRT}{PV}$ ⇒ since the mass (m) is in the numerator the M_m is directly proportional to the mass. Therefore, if the mass is smaller than it should be due to the leak, the final ~~Calculated M_m will also be smaller then the true value.~~

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Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 6

Empirical Formula

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 19/4/2026

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(16) درجة

Data and Calculation

DATA	Value	*Unit*
Mass of empty crucible (m_1)	12.94	g
Mass of crucible + Magnesium (m_2)	13.20	g
Mass of crucible + Magnesium Oxide (m_3)	13.31	g
Mass of Magnesium Oxide $m_{Mg\,x\,O_Y}$ (m_3-m_1)	0.37	g
Mass of Magnesium m_{Mg} (m_2 - m_1)	0.26	g
Moles of Magnesium $X = m_{Mg} \backslash 24.3$ (g\mol)	$\frac{0.26}{24.3} = 0.0107$.	mol
Mass of Oxygen $m_O = m_{(Mg\,X\,OY)} - m_{Mg}$	0.11	g
Moles of Oxygen $Y = m_O \backslash 16$(g\mol)	$\frac{0.0107}{0.069} = 1.55 * 1$ 0.00687	
The mole ratio of Magnesium to Oxygen X:Y	$1.5 * 2 = 3$ / 3 : 2	
The Empirical formula of Magnesium oxide	Mg_3O_2	

M_2O ✗

Show Calculation:

1- Moles of magnesium =

$n_{Mg} = \frac{mass\ mg}{M \cdot W_{Mg}} \Rightarrow \frac{0.26}{24.3} \Rightarrow 0.0107\ mol$

2- Moles of oxygen =

$n_o = \frac{mass\ O}{M \cdot W_O} \Rightarrow \frac{0.11}{16} \approx 0.00687\ mol$

3- The Mole ratio =

$\frac{n_{Mg}}{n_O} : \frac{n_O}{n_O} \Rightarrow \frac{0.0107}{0.00687} : \frac{0.0687}{0.0687}$

$2 * (1.55 : 1)$

$(3 : 2)$

Note: This discrepancy in the results is due to experimental errors, the actual empirical formula for magnesium oxide is: M_2O

Questions:

1) How would the calculated molar ratio between (Mg/O) be affect if:

a- Forget the addition of water to get rid of magnesium nitride form.

Effect: The calculated ratio will be higher than the theoratical value

explanation: Mg reacts with (N_2) in the air to from Mg_3N_2. since(N) has a lower M_m (14)g/mol than(O)→16g/mol the nitride byproduct weighs less than the(O) This results is low calcuated mass for(O) an overall higher caculated value

b- Incomplete dehydration was done after water was added.

Effect: the caculated ratio will be lower than the theoretical value

explation: Residual water adds to the final mass of the product this excess weight is incorrectly attributed to oxygen a larger mass of oxygen in the denominator decreases the overall MgO ratio

c- Some of magnesium oxide spilled out before final weighting.

Effect: the calculated ratio will be higher than the real value

explation: spilling the product leads to a lower final mass reading. When calculating the mass of oxygen (final mass - Initial Mg mass), the value will be erroneously small. A very small value for oxygen in the denominator significantly inflates the MgO ratio

2) Calculate Mg% of your compound in this experiment.

$$Mg\% = \left[\frac{\text{Mass of Mg}}{\text{Mass of MgOxide}}\right] * 100.0\%$$

$$= \left[\frac{13.20 - 12.94}{13.31 - 12.94}\right] * 100\% \Rightarrow \left[\frac{0.26}{0.37}\right] * 100\% \Rightarrow 70.27\%$$

- conclusion: -

* Empirical formula: The experiment aims to find the simplest number ratio of doms in the magnesium oxide compound.
* Applying the law of conservation of mass: the mass of reacted oxygen was determined by subtracting the inital mass of magnesium from the final product mass
* Molar Analysis: Measured masses and molar masses (Mg=24, O=16) were used to convert weight data into moles, enabling the comparison of atomic ratios.

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Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 7

Precipitation Reaction (Limiting Reactant)

Student Name: [redacted]

Student Number: [redacted]6

Submission Date: 26/4/2026

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$1CaCl_2 + 1Na_2CO_3 \longrightarrow 1CaCO_3 + 2NaCl$

(7) نسبة

Data and calculations:

* Determination of the limiting reactant: $m_2 = 1.23$

Data		Unit
Volume of $CaCl_2$	15	mL
Molarity of $CaCl_2$	0.5	molar
Volume of Na_2CO_3	15	mL
Molarity of Na_2CO_3	1.5	molar
Number of moles of $CaCl_2$ reacted ($M \times V_L$)	0.0075	mol
Number of moles of Na_2CO_3 reacted ($M \times V_L$)	0.0225	mol
The Limiting reactant is	$CaCl_2$	–
The Excess reactant is	Na_2CO_3	–
The mass of $CaCO_3$ expected (Theoretical yield)	0.0075 * 100.09 = 0.750	g
The mass of $CaCO_3$ precipitate (Actual yield)	$m_2 - m_1$ = 0.64	g
% Yield of $CaCO_3$ = $\frac{Actual\ Yield}{Theoretical\ Yield} \times 100\%$	$\frac{0.64}{0.750} * 100\% = 85.33$	%
mass of Filter paper (m_1)	0.59	g

لإيجاد عدد مولات الناتج

0.069

Show calculations

Moles of $CaCl_2$	Moles of Na_2CO_3
$n = M * V_L = 0.5 * 0.015$ $= 0.0075$ mol	$n = M * V_L = 1.5 * 0.015$ $= 0.0225$ mol
Moles of $CaCO_3$ expected	**% Yield of $CaCO_3$**
$n\ CaCl_2 = 0.0075$	$\frac{actual}{theoretical} * 100\% \Rightarrow \frac{0.64}{0.750} * 100\%$ $\Rightarrow 85.33\%$

mins to ml's

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Questions:

1) If 23 grams of iron (II) chloride ($FeCl_2$) reacts with 41 grams of sodium phosphate (Na_3PO_4), what is the limiting reagent? How much sodium chloride can be formed?

$$3FeCl_2 + 2Na_3PO_4 \longrightarrow Fe_3(PO_4)_2 + 6NaCl$$

$n(FeCl_2) = \frac{23}{126.75} \approx 0.1815 mol$ // $n(Na_3PO_4) = \frac{41}{163.94} = 0.250 1 mol$

$\frac{0.18}{3} \Rightarrow 0.06 mol$ // $\frac{0.25}{2} = 0.125 mol$

L.R excess

$FeCl_2$ $NaCl$

3mol : 6mol

0.18 mol : X

$\Rightarrow X = \frac{0.18 * 6}{3} \Rightarrow 0.36 mol$

- mass of $NaCl = 0.36 * 58.44 = 21.06 g$

2) How much of the excess reagent remains when this reaction has gone to completion?

$FeCl_2$ Na_3PO_4

3mol : 2mol

0.18 : Z mol

$\Rightarrow Z = \frac{0.18 * 2}{3} = 0.12 mol$

$0.25 - 0.12 = 0.129 * 0.13 mol$

mass remains $\Rightarrow n * Mm = 0.13 * 163.94 \Rightarrow 21.15 g$

Conclusion:

This experment demonstrated that chemical reactant, which determins the maximum amount of product that can be formed. I also learned to distingwish between theorical yield serves as a key indicator of experiment's efficiency and accuracy. ~~Understanding efficiency and accuracy~~ Understanding these consepts and essential for optimizing reaction and minimizing waste in practical applications.

10/10 3/5

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Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 8

Volumetric Analysis I (Vinegar Analysis)

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 3/5/2026

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تجربة (8)

Data and Calculations

		Unit
Volume of diluted vinegar solution	10	mL
Molarity of NaOH solution	0.1	mol/L
Initial burette reading of NaOH solution	3.9	mL
Final burette reading of NaOH solution	17.3	mL
Volume of NaOH solution	13.4	mL
Moles of NaOH	$\frac{13.4}{1000} * 0.1 \Rightarrow 1.34 * 10^{-3}$	mol
Moles of CH_3COOH	1.34×10^{-3}	mol
Mass of CH_3COOH	0.0812	g
Molarity of diluted vinegar solution	$10 * 1.07 \Rightarrow 10.7$	
Mass percentage of CH_3COOH in solution ($\%CH_3COOH$)		

Show calculations:

$\frac{0.0812}{10.7} \times 100\% \Rightarrow \simeq 7.58 \times 10^{-3}$

0.75 %

$\%100 * \frac{0.0812}{1.07} \simeq 7.5\%$

(A) 26M

to find the molas of NaOH $\Rightarrow \frac{13.4}{1000} * 0.1 \Rightarrow 1.34 \times 10^{-3}$

(moles of NaOH = mols of CH_3COOH)

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QUESTIONS

1. **What was the purpose of the phenolphthalein indicator in this experiment?**

The purpose of phenolphatein is to act as an indicator to determine the end point of titration, it change color from colorless to faint pink color

2. **Suppose you added 40 mL of water to your vinegar sample instead of 20 mL. Would the titration have required more, less or the same amount of NaOH solution for a complete reaction? Explain.**

the same amount of NaOH, beause the titation depends on the number of moles of acetic acid present in sample adding more distled water changes the concentration (molarty) and the volume but not chang the total number of moles of acid

3. **Suppose you had titrated your vinegar sample with barium hydroxide instead of sodium hydroxide:** $Ba(OH)_{2(aq)} + 2CH_3COOH_{(aq)} \rightarrow Ba(CH_3COO)_{2(aq)} + 2H_2O_{(l)}$
Consider a 0.586 M aqueous solution of barium hydroxide. What volume (in mL) of 0.586 M $Ba(OH)_2$ solution are required to neutralize 10 ml of vinegarcontaining 2.78 g of acetic acid?

1mole of Ba : 2mol of $CH_3COOH \Rightarrow (1:2)$

* $n_{vinger} = \frac{2.78}{60.05} \Rightarrow 0.0463 \Rightarrow \frac{0.0463}{2} \Rightarrow 0.02315$ mol

* $V = \frac{0.02315}{0.586} = 0.03951 L \Rightarrow 39.5$ mL

Conclusion:- In this experment we used acid-base titration to analyze vineger. We learned the phenolphthalein is essential to know the end point, and when we adding water (dilution) does not change the titration result becaues the number of moles of acid is constant.

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Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 9

Volumetric Analysis II (Bleach Analysis)

Student Name: [redacted]

Student Number: [redacted]

Submission Date: 10/5/2026

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تجربة (٩)

Data and calculation:

DATA	Trial 1	Trial 2	Unit
Volume of diluted bleach solution	10		mL
Dilution factor	10		
Molarity of $S_2O_3^{-2}$ solution	0.1		M
Initial burette reading of $S_2O_3^{-2}$ solution	12		mL
Final burette reading of $S_2O_3^{-2}$ solution	22.4		mL
Volume of $S_2O_3^{-2}$ solution	10.4		
Moles of $S_2O_3^{-2}$	0.00104		
Moles of I_2	0.00052		
Moles of OCl^-	0.00052		
Molarity of diluted bleach	0.052		
Molarity of original concentrated bleach	0.52		

٨ 3.85

A) Mass percentage of NaOCl in original bleach solution (%NaOCl). *(Show your calculations)*

mass ⇒ 0.52 * 74.5 ⇒ 38.71

% ⇒ 38.71 / 1080 * 100% ⇒ 3.85%

B) Mass percentage of Cl_2 in original bleach solution (%Cl_2). *(Show your calculations)*

mass ⇒ 0.5 * 70.9 = 36.87

Mass ⇒ D * 1000 = 1080

%Cl_2 ⇒ 36.87 / 1080 * 100% ⇒ 3.41%

-2

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- leads to the number of moles of iodine which in turns allows the determination of the moles of hypochlorite in the diluted bleach.
- Finally, the number of hypochlorite moles in the original bleach

$n(S_2O_3^{-2}) = M * V \Rightarrow 0.10 \times 0.0104 = 0.00104\ mol$

$n I_2 \Rightarrow \frac{0.00104}{2} = 0.0005\ mol$

the ratio between I_2 and OCl^-

OCl^- in diluted bleach $= 0.00052\ mol$

in the original bleach $\Rightarrow 0.00052 \times 10 \Rightarrow 0.0052$

☆

In this experiment, the concentration of hypochorite in bleach was determined by iodometric titration using thiosulfate. The moles of iodine and hypochlorite then the dilution factor was applied to find the concentration of the original bleach solution and its mass percent.

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Al-Balqa Applied University

Faculty of Engineering Technology

Scientific Basic Sciences Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 10

Thermochemistry and Calorimetry

Student Name:

Student Number:

Submission Date:

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10/10 10/5 (10) نصيحة

Data & Calculation

Part 1:

DATA	Value	Unit
Mass of empty cup-calorimeter m_1	9.45	g
Mass of empty cup-calorimeter + $MgCl_2$ solution m_2	33.27	g
Mass of $MgCl_2$ solution (m_2-m_1)	23.82	
T_i	24	C°
T_f	30	C°
ΔT	6	C°
Heat gained by the calorimeter $q_{cal.}$	Zero	
Heat gained by the solution $q_{sol.}$ $C \times m \times \Delta T$ (4.2)	597.8	
Heat lost by the reaction $q_{rxn.}$	- 597.8	
Mass of Mg m_{Mg}	0.1	
Number of moles of Mg n_{Mg} $\frac{0.1}{24}$	0.0041	
ΔH_1 $\Rightarrow \frac{-597.8}{0.0041}$	-145.8	KJ/mol

Part 2:

DATA	Value	Unit
Mass of empty cup-calorimeter (m_1)	9.39	g
Mass of empty cup-calorimeter + $MgCl_2$ (m_2)	33.43	g
Mass of $MgCl_2$ solution (m_2-m_1)	24.04	
T_i	24	C°
T_f	28	C°
ΔT	4	C°

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Heat gained by the calorimeter	q_{cal}	0	c°
Heat gained by the solution $c * m * \Delta t$	q_{sol}	403.8	
Heat lost by the reaction	q_{rxn}	-403.8	
Mass of MgO		0.21	g
Number of moles of MgO $\frac{0.21}{40}$	n_{MgO}	0.00525	mol
$\Delta H°_2$ $\frac{-403.8}{0.0052}$		-77.5	kJ/mol

Calculation of $\Delta H_f°$

$\Delta H°_{form} = \Delta H°_1 + (-\Delta H°_2) + \Delta H°_3$

$\Delta H°_{form.} = -145.8 + 77.5 + -286 kJ = -354.3$

kJ

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